



Federal Urdu University
of Arts, Science and Technology

ARTIFICIAL INTELLIGENCE

AI Agent Assignment

Agent : Smart Umbrella Agent

Submitted by : Syeda Haseeba Noor

Father's Name : Syed Hamid Raza Shah

Roll Number : 23122116

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1.Agent Name : Smart Umbrella Agent

The **Smart Umbrella Agent** is a proposed AI-based agent that helps a user deal with changing weather conditions, especially rain. The agent collects information about the surrounding environment, such as rain, temperature, humidity, wind conditions, and weather forecasts. After analyzing this information, it decides whether the user may need an umbrella and provides an appropriate response.

For example

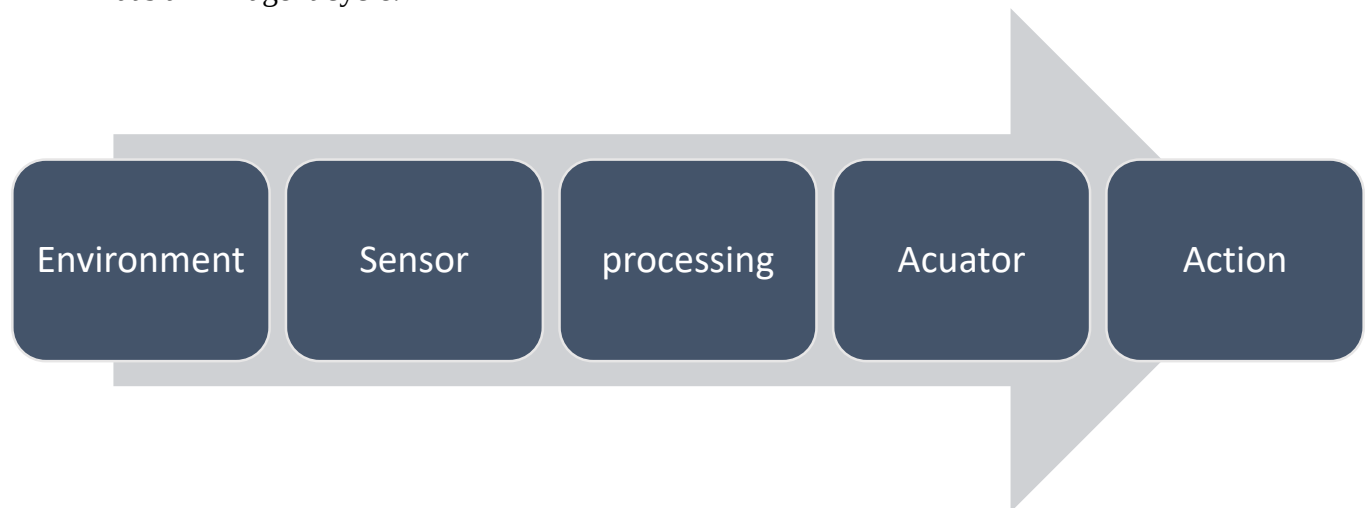
If the user is about to leave home and the system detects a high possibility of rain, the Smart Umbrella Agent can give an alert or notification to remind the user to take an umbrella. If there is no significant possibility of rain, the agent does not need to give a rain warning.

The main purpose of the Smart Umbrella Agent is to make a simple everyday task more convenient by observing the environment and making a useful decision based on the information it receives, real-life problem.

2. Graphical Representation of Smart Umbrella Agent

An AI agent observes information from its environment, processes that information, makes a decision, and performs an action to achieve a goal.

Thin That's an AI-agent cycle.



- Environment = Rain, Sun and Wind
- Sensor = Rain sensor detects rain, light sensor
- Processing = The agent decides that protection is required
- Actuator = Opening operators, closing
- Action = Umbrella opens , close

That's an AI agent .

3. Agent Formulation

Component	Smart Umbrella
Agent	Smart Umbrella
Environment	Outdoor surroundings, rain, sunlight, wind, user
Sensors	Rain sensor, light sensor, optionally wind sensor
State	Rain condition, umbrella status, light condition, wind condition
Processing	Analyzes sensor information and determines the required action
Actuators	Opening and closing mechanism/motor
Goal	Provide appropriate protection to the user

4.Environmental Types

Environment Classification

- **Partially Observable:** Sensors cannot observe everything.
- **Stochastic:** Weather is uncertain.
- **Sequential:** Current decisions can affect future states.
- **Dynamic:** Weather can change over time.
- **Continuous:** Weather values such as temperature and humidity change continuously.
- **Single Agent:** The umbrella operates as one agent.
- **Partly Known:** The agent knows some conditions but cannot know future weather with certainty.

For example:

Dynamic

Weather can change while the umbrella is operating.

Stochastic

Weather conditions cannot be predicted with complete certainty.

Partially observable

The umbrella's sensors cannot observe every aspect of the surrounding world.

5. PEAS Framework

- P = Performance Measure
- E = Environment
- A = Actuators
- S = Sensors

PEAS	Smart Umbrella
Performance Measure	Protect the user appropriately and respond quickly to weather changes
Environment	Outdoor surroundings, rain, sunlight, wind and user
Actuators	Opening/closing mechanism
Sensors	Rain sensor, light sensor and optionally wind sensor

Working Example

Rain begins



Rain sensor detects rain



Agent receives the information



Current state is checked



Agent decides to open



Opening actuator operates



Umbrella becomes OPEN